Multi-Soliton Wave Trains in the KdV Equation:

A Preliminary PINN Study

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Abstract

Building on our earlier single-soliton PINN work, we extend the method to a three-soliton wave train modelling the propagating pattern left behind when a moving disturbance (e.g., a boat) stops in a shallow canal. We switch to the positive-amplitude KdV convention and place three solitons of speeds $c = 2.5, 2.0, 1.5$ in a configuration that avoids collisions within the time window. Using essentially the same 6×50 architecture as our single-soliton baseline, we achieve a relative L^2 error of 0.18% on the full $[-20, 40] \times [0, 5]$ domain.

Contents

1 Motivation

John Scott Russell's 1834 observation was not of a single soliton but of a *train* of them. When the boat he was following abruptly stopped in the Union Canal, the displaced water separated into a procession of solitary waves, each maintaining its shape as it travelled. This is the natural starting point for any realistic application of the KdV equation: real disturbances produce multiple waves, not one.

Our earlier work established a PINN baseline on the single-soliton problem and ran a series of ablations (noise robustness, collocation density, initial-condition substitution, wave-speed sensitivity, data-point sweep). Those results gave us a working understanding of when PINNs succeed and when they break. The next question is whether that understanding transfers to the multi-wave case.

This document reports a preliminary three-soliton result. The goal is not a finished study but a proof-of-concept that the single-soliton architecture scales to a wave-train scenario without major modifications. We also collect a set of open questions that we would like to discuss with Prof. Saha before planning the full SOP.

2 Problem Setup

2.1 The KdV Equation (Positive Convention)

In our single-soliton paper we used the negative-amplitude convention because it matched the reference text we were working from [1]. For this extension we switch to the standard positive-amplitude form:

$$\frac{\partial u}{\partial t} + 6u\,\frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0, \qquad (x,t) \in \Omega = [-20,\, 40] \times [0,\, 5]. \tag{1}$$

This is the form in which solitons appear as peaks rather than troughs, which is easier to visualise for a wave-train configuration. The two conventions are related by $u \mapsto -u$ and are physically equivalent.

2.2 Single-Soliton Building Block

The positive-convention single-soliton solution of Eq. (1) is

$$u_c(x,t;\, x_0) = \frac{c/2}{\cosh^2\!\left(\frac{\sqrt{c}}{2}\,(x - ct - x_0)\right)}, \tag{2}$$

where c is the wave speed and x_0 is the initial centre. The peak amplitude is $c/2$ and the soliton is located at $x = ct + x_0$ at time t.

2.3 Three-Soliton Ansatz

KdV admits exact multi-soliton solutions via the Hirota bilinear method, but for well-separated solitons whose profiles do not overlap, a simple superposition of single-soliton profiles is an accurate approximation. We use

$$u(x,t) = u_{2.5}(x,t;\, 10) + u_{2.0}(x,t;\, 0) + u_{1.5}(x,t;\, -10). \tag{3}$$

The speeds and positions are chosen so that:

- The fastest wave is in front (rightmost), the slowest at the back. Since taller solitons travel faster, this ordering prevents collisions within the time window.
- The initial separation of 10 units between centres is large enough that the individual sech^2 profiles have decayed to $\sim 10^{-4}$ at the overlap region, so the superposition error is negligible at $t = 0$.

The peak amplitudes at $t = 0$ are 1.25, 1.0, and 0.75. By $t = 5$, the fastest soliton has travelled to $x = 22.5$, the middle one to $x = 10$, and the slowest to $x = -2.5$. The spacing between the outermost waves grows from 20 to 25 units; within the time window, each pair remains well-separated throughout.

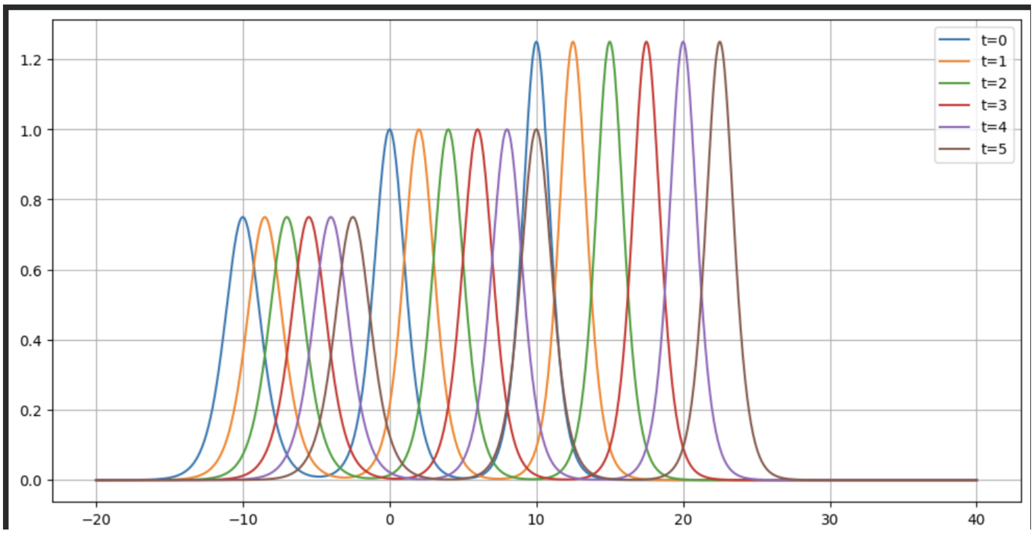

Figure 1: Exact three-soliton wave train at $t = 0, 1, 2, 3, 4, 5$. The fastest soliton (rightmost) maintains a lead; the slowest (leftmost) falls behind but none of the waves catch each other within the domain.

2.4 Initial and Boundary Conditions

The initial condition is the wave train at $t = 0$:

$$u(x, 0) = u_{2.5}(x, 0;\, 10) + u_{2.0}(x, 0;\, 0) + u_{1.5}(x, 0;\, -10). \tag{4}$$

All three solitons decay exponentially away from their centres, so the boundaries are effectively zero:

$$u(-20,\, t) \approx 0, \qquad u(40,\, t) \approx 0, \qquad t \in [0, 5]. \tag{5}$$

The domain is wider than in the single-soliton case because three waves need more room.

3 Network and Training

3.1 Architecture

We reuse the single-soliton architecture: a fully connected feedforward network with 2 inputs (x, t), six hidden layers of 50 neurons each with tanh activation, and 1 output $u_\theta(x, t)$. Weights are Xavier-initialised; biases are zero.

We deliberately kept the architecture identical to test whether the single-soliton recipe transfers without modification. A wider or deeper network may be necessary for more demanding multi-soliton configurations; this is one of the open questions we want to address in the SOP.

3.2 Loss and Sampling

The loss structure is the same as the single-soliton baseline with all four terms active:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{data}}. \tag{6}$$

The PDE residual uses the positive convention:

$$r(x,t) = \frac{\partial u_\theta}{\partial t} + 6u_\theta \frac{\partial u_\theta}{\partial x} + \frac{\partial^3 u_\theta}{\partial x^3}. \tag{7}$$

We scale the number of points with the larger domain:

- $N_r = 20{,}000$ collocation points (vs. 10,000 before), since the domain area doubled.
- $N_{\text{data}} = 400$ scattered measurements (vs. 200 before), same reason.
- $N_{\text{IC}} = 500$ points along $t = 0$ (unchanged).
- $N_{\text{BC}} = 100$ points per edge at $x = -20$ and $x = 40$ (unchanged).

3.3 Training Schedule

Identical to the single-soliton baseline: Adam for 15,000 iterations at lr $= 10^{-3}$, followed by L-BFGS for 2,000 iterations at lr $= 0.1$, `max_iter` $= 20$, `history_size` $= 50$.

4 Results

4.1 Error

After Adam alone: L^2 relative error of 2.31%. After L-BFGS fine-tuning: $L^2 = 0.18\%$.

L-BFGS provides a significant improvement ($> 10\times$), consistent with our single-soliton observation that L-BFGS helps under clean data. The gradient-conflict finding from the noise experiment does not apply here because the data is noise-free.

4.2 Solution Quality

Figure 2 shows the PINN prediction overlaid on the exact wave train at six time steps. The two are visually indistinguishable; all three solitons are tracked through the full time window.

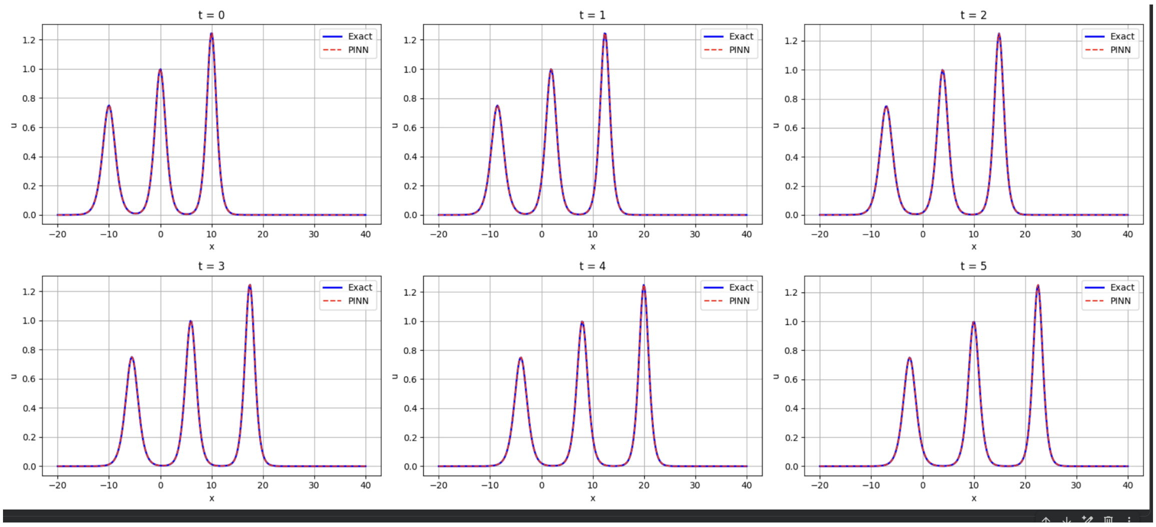

Figure 2: PINN prediction (red dashed) vs. exact three-soliton wave train (blue solid) at $t = 0, 1, \ldots, 5$. All three waves are captured accurately; no visible drift or amplitude decay.

4.3 Error Distribution

The pointwise absolute error (Figure 3) shows three diagonal bands, one per soliton, with error concentrated along each wave's trajectory. Maximum pointwise error is ~ 0.006, and no single soliton dominates the error budget—all three are resolved comparably. Error grows slightly at later times, matching the accumulation pattern seen in the single-soliton case.

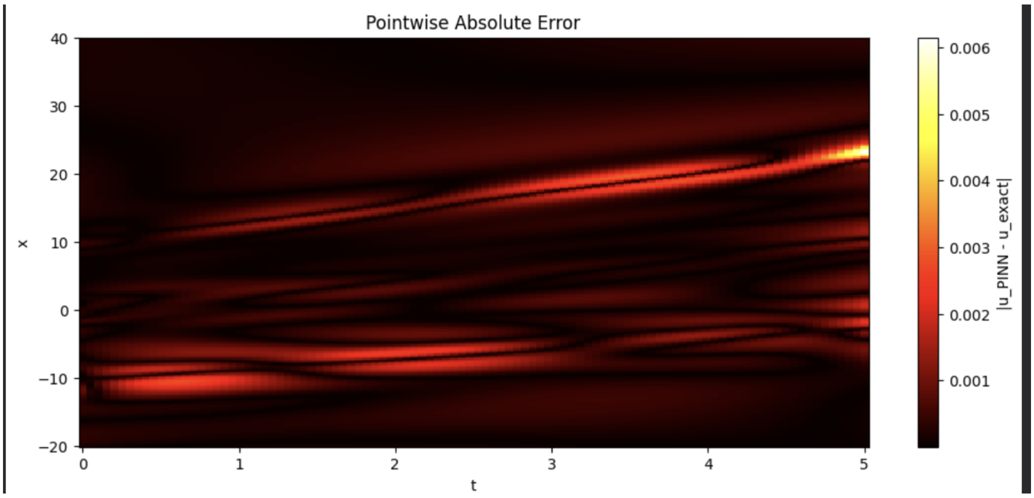

Figure 3: Pointwise absolute error $|u_\theta - u_{\text{exact}}|$. Three diagonal bands correspond to the three soliton paths. Error grows mildly with t.

5 Discussion

The key observation from this preliminary run is that **the single-soliton architecture transfers to the three-soliton case with only trivial modifications**: the domain widened, the PDE sign flipped, and the ground truth function was replaced. No architectural changes, no new loss terms, no hyperparameter retuning. The final error (0.18%) is within a factor of 3 of the single-soliton baseline (0.07%), which is a reasonable penalty for handling three waves instead of one.

This is encouraging, but it is only one data point. We have not yet tested whether the findings from our single-soliton ablations (noise robustness, collocation saturation, IC substitution, data-point saturation) carry over to the multi-wave case. In particular, the IC substitution result—that scattered data can replace the initial condition—is unlikely to hold here, because a multi-soliton IC encodes more information (number of waves, individual amplitudes, positions) than a handful of data points can easily communicate.

The three-soliton train is also a relatively easy multi-wave configuration. The solitons are well separated, so the superposition ansatz is effectively exact. Harder configurations—solitons close enough to interact, or a train of 5–10 waves—will likely require the true Hirota solution as ground truth and may stress the current architecture.

6 Conclusion

A preliminary PINN extension to a three-soliton wave train achieves $L^2 = 0.18\%$ on the $[-20, 40] \times [0, 5]$ domain using the same architecture as our single-soliton baseline. The result demonstrates that the method transfers without major modification to a more realistic multi-wave scenario.

References

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[4] R. Hirota, "Exact solution of the Korteweg–de Vries equation for multiple collisions of solitons," *Phys. Rev. Lett.*, vol. 27, pp. 1192–1194, 1971.